

Functional Requirements Document (FRD)

Health Risk Dashboard

TY B.Sc. Data Science – Mini Project

1. Purpose

This document defines the functional requirements for the Health Risk Dashboard. The system will analyze anonymized healthcare data, visualize health-risk patterns, and provide a basic machine-learning risk prediction with explainable results.

2. Scope

The system will support healthcare dataset upload, data preprocessing, exploratory analysis, interactive dashboard visualization, health-risk prediction, and explanation of model results. The project is intended for academic and analytical use and is not a medical diagnosis system.

3. Target Users

- Project students/developers – build and maintain the dashboard.
- Faculty/project evaluators – review functionality, analytics, and model performance.
- Academic users – explore healthcare trends and risk factors.

4. Functional Requirements

FR-01: The system shall allow the user to upload an approved healthcare CSV dataset.

FR-02: The system shall validate required columns and report missing or invalid fields.

FR-03: The system shall detect and handle missing values and duplicate records.

FR-04: The system shall perform basic data preprocessing and feature preparation.

FR-05: The system shall display total patients, risk counts, and key health indicators.

FR-06: The dashboard shall provide filters for age group, gender, risk category, and available lifestyle/medical attributes.

FR-07: The dashboard shall visualize relationships between risk and factors such as BMI, blood pressure, glucose, cholesterol, smoking, and physical activity where available.

FR-08: The system shall accept patient attributes through a prediction form.

FR-09: The system shall use a trained classification model to predict the selected health-risk category.

FR-10: The system shall display prediction confidence/probability where supported by the model.

FR-11: The system shall provide feature-importance or SHAP-based explanations for predictions.

FR-12: The system shall display model evaluation results including accuracy, precision, recall, F1-score, and confusion matrix.

FR-13: The system may provide a simple natural-language explanation using a locally hosted open-source LLM; this must not provide medical diagnosis or treatment advice.

5. Non-Functional Requirements

Usability: Dashboard controls and charts should be understandable to a college-level user.

Performance: Normal dashboard interactions should respond within a few seconds on the target laptop.

Privacy: Only anonymized/academic datasets shall be used; personally identifiable information shall not be required.

Reliability: Invalid files or missing fields shall produce clear error messages.

Maintainability: Code shall be organized into separate modules for preprocessing, modeling, explanation, and UI.

Portability: The core application should run locally on a standard student laptop.

6. System Workflow

1. User uploads healthcare dataset.
2. System validates and preprocesses the data.
3. Cleaned data is analyzed for descriptive statistics and patterns.
4. Dashboard displays KPIs, charts, and interactive filters.
5. User enters patient attributes for prediction.
6. ML model predicts the selected health-risk category.
7. SHAP/feature importance identifies influential factors.
8. Dashboard displays prediction and explanation.
9. User reviews model performance and analytical insights.

7. Recommended Technology Stack

Frontend/UI: Streamlit

Backend: Python

Data Processing: Pandas, NumPy

Visualization: Plotly, Matplotlib, Seaborn

Database: SQLite (optional for MVP; CSV is sufficient initially)

Machine Learning: Scikit-learn Random Forest Classifier

Explainable AI: SHAP

Optional local LLM: Qwen3-4B through Ollama

Development: Jupyter Notebook or VS Code

Version Control: Git and GitHub

Deployment: Local laptop; optional free hosting for the dashboard where compatible

8. Dataset Requirements

Recommended attributes: Patient ID, Age, Gender, BMI, Blood Pressure, Glucose Level, Cholesterol, Smoking Status, Physical Activity, Medical History, and Health Risk/Outcome.

Recommended dataset size: approximately 1,000–10,000 records for the MVP.

Sources may include UCI Machine Learning Repository, Kaggle datasets with suitable licenses, and government/public datasets.

Data must be anonymized and legally/academically permitted for use.

9. Machine Learning Requirements

Primary algorithm: Random Forest Classifier.

Baseline algorithms: Logistic Regression and Decision Tree.

Optional advanced algorithms: Gradient Boosting/XGBoost or a small neural network if time and hardware permit.

Train/test split and/or cross-validation shall be used to evaluate generalization.

Class imbalance shall be checked and addressed where necessary.

10. Explainable AI Requirements

SHAP shall be used to explain individual predictions and identify globally important features.

Explanations shall clearly distinguish model association from medical causation.

The dashboard shall show the top contributing features in a simple visual format.

11. Evaluation Metrics

Classification: Accuracy, Precision, Recall, F1-score, Confusion Matrix, and ROC-AUC where applicable.

Dashboard: calculation correctness, filter functionality, readability, and response time.

XAI: consistency of explanations with the trained model and correct mapping of feature contributions.

Optional LLM: factual consistency with computed results and absence of unsupported medical claims.

12. MVP Requirements

- One approved healthcare dataset
- Data cleaning and preprocessing
- Interactive dashboard with at least 5–7 useful visualizations
- Risk-category prediction
- Random Forest model
- SHAP explanation
- Patient prediction input form
- Model evaluation page
- GitHub repository with documentation

13. Future Enhancements

Real-time data integration; personalized risk profiles; additional ML/deep-learning models; local LLM-generated summaries; voice interaction; mobile-friendly UI; additional XAI methods; and integration with approved healthcare information systems subject to privacy and security requirements.

14. Constraints & Assumptions

The project assumes availability of a suitable anonymized dataset and a laptop capable of running Python and the selected models. The dashboard is an educational analytics system and must not be used as a substitute for professional medical diagnosis or treatment.

15. Acceptance Criteria

The project will be considered functionally complete when a user can upload a valid dataset, view correct dashboard KPIs and visualizations, filter the data, submit patient attributes, receive a risk prediction, view an explanation of the prediction, and review model evaluation metrics without application errors.